Multi-Agent Case Report

Case: kg_convergence_demo

Contributing agents: ecg, mri, patient

Note: synthetic 3-modality pairing for architecture demonstration - the ECG, MRI, and patient-data records are not from the same real patient.

Integrated Summary (Sep Agent)

The ECG shows a Sokolow-Lyon voltage index exceeding the threshold, suggesting left ventricular hypertrophy (LHV). The MRI indicates a hypertrophic cardiomyopathy pattern. Resting blood pressure is elevated. These findings suggest hypertensive heart disease.

Flags for review: high_resting_bp_suspected, lvh_sokolow_lyon_suspected, pathology_HCM_pattern_suspected

Cross-Modality Convergent Evidence (Knowledge Graph)

hypertensive heart disease - independently supported by ecg, mri, patient (high_resting_bp_suspected, lvh_sokolow_lyon_suspected, pathology_HCM_pattern_suspected)
left ventricular hypertrophy - independently supported by ecg, mri (lvh_sokolow_lyon_suspected, pathology_HCM_pattern_suspected)
Convergence across independent modalities is stronger evidence than any single modality. This block is computed deterministically from the clinical knowledge graph, not generated by the language model.

Traceability

Statement	Agent	Feature
Sokolow-Lyon voltage index exceeds 3.5 mV	ecg	sokolow_lyon_index_mv
Heart rate is within normal range	ecg	heart_rate_mean_bpm
Pathology classifier indicates hypertrophic cardiomyopathy pattern	mri	pathology_HCM_probability
Resting blood pressure is elevated	patient	resting_bp_mmHg

Note: traceability confirms each statement cites a real finding from a contributing agent - it does not verify that the summary's characterization of that finding (e.g. "low" vs "high") is clinically correct.

ECG Agent Findings

Record: synthetic_lvh_ecg

Flags: lvh_sokolow_lyon_suspected

Feature	Value	Confidence	Evidence type	Rationale
sokolow_lyon_index_mv	4.20 mV	0.80	numeric_feature	Sokolow-Lyon voltage index exceeds the 3.5 mV threshold, consistent with left ventricular hypertrophy.
heart_rate_mean_bpm	78 bpm	0.90	numeric_feature	Heart rate within normal range.

MRI Agent Findings

Record: synthetic_hcm_mri

Flags: pathology_HCM_pattern_suspected

Feature	Value	Confidence	Evidence type	Rationale
pathology_HCM_probability	0.95	0.90	numeric_feature	Learned pathology classifier indicates a hypertrophic cardiomyopathy pattern on the raw MRI.

PATIENT Agent Findings

Record: synthetic_htn_patient

Flags: high_resting_bp_suspected

Feature	Value	Confidence	Evidence type	Rationale
resting_bp_mmHg	160 mmHg	0.75	numeric_feature	Elevated resting blood pressure.